

Business Impact Analysis (BIA) Report

Target Organization: ApexPay Solutions Inc.

Industry: Financial Technology (FinTech)

Prepared By: Divine Echezona Smart, Junior Cyber Risk Analyst

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1. Executive Summary

This Business Impact Analysis evaluates the potential operational, financial, and regulatory impacts resulting from disruptions to core business operations at ApexPay Solutions. The objective is to identify critical business functions, establish recovery priorities (**RTO** and **RPO**), and guide resilient architecture planning.

2. Business Function Criticality Framework

To establish clear recovery priorities during a disaster, business processes are categorized into four criticality tiers:

© Tier 1: Mission-Critical (RTO ≤ 2 Hours): Failure results in immediate financial loss, regulatory violations, or severe brand destruction.

© **Tier 2: Business-Critical (RTO 4–12 Hours):** Severe operational friction occurs, but short-term workarounds exist.

© **Tier 3: Important (RTO 24–48 Hours):** Causes minor operational delays with minimal immediate financial exposure.

© **Tier 4: Non-Critical (RTO \$>\$ 72 Hours):** Internal support functions that can remain offline for several days without business disruption.

3. Impact Assessment & Recovery Targets

Business Process	Criticality Tier	RTO Target	RPO Target	Financial Impact (Per Hour)	Primary Non-Financial Impact
API Payment Processing	Tier 1 (Mission-Critical)	15 mins	0 mins (Near real-time)	~\$45,000 / hr	Immediate customer defection, merchant SLA breaches.
User Authentication / Identity	Tier 1 (Mission-Critical)	1 hour	5 mins	~\$25,000 / hr	Complete lockout for all merchant dashboards.
Merchant Settlement & Payouts	Tier 2 (Business-Critical)	8 hours	1 hour	~\$5,000 / hr	Merchant dissatisfaction, delayed payouts, regulatory inquiries.
Fraud Detection Engine	Tier 2 (Business-Critical)	4 hours	15 mins	~\$15,000 / hr	Higher fraud risk exposure during bypass mode.
Customer Support Ticketing	Tier 3 (Important)	24 hours	12 hours	Minimal direct loss	Operational backlog, customer frustration.
Internal HR & Payroll	Tier 4 (Non-Critical)	72 hours	24 hours	\$0 / hr short-term	Delayed internal administrative tasks.

4. System & Resource Dependencies Mapping

For a business process to recover within its assigned RTO, all supporting dependencies must share equal or strict recovery metrics.

Primary Asset: Core Payment Processing API

© **Dependencies:**

- *Cloud Infrastructure:* AWS US-East-1 (Primary) & AWS US-West-2 (Failover).
- *Database:* PostgreSQL RDS Multi-AZ Deployment.
- *Third-Party Vendors:* Visa/Mastercard Network Connectors.
- *Internal Team:* Cloud DevOps & On-Call SRE Team.

© **Single Point of Failure (SPOF) Identified:** Primary database write-node currently lacks automatic multi-region failover, threatening the 15-minute RTO target in the event of a regional AWS outage.

5. Risk-Prioritized Remediation Plan

1. Implement Multi-Region Database Auto-Failover:

Finding: The current RPO/RTO targets for payment processing cannot be guaranteed under regional cloud failure.

Recommendation: Deploy AWS Aurora Global Database to achieve cross-region storage replication under 1 second.

2. Establish Automated Third-Party Health Checks:

Finding: Dependency on credit card networks lacks real-time automated fallback.

Recommendation: Implement multi-acquirer routing to automatically bypass degraded card processors.

3. Formalize Standard Operating Procedures (SOPs):

Finding: Manual intervention is required to fail over the authentication service.

Recommendation: Script automated failover triggers via Infrastructure-as-Code (Terraform/Ansible).